

Structural and Excitation-Dependent Practical Identifiability in Lumped Thermal Parameter Estimation

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Abstract

Lumped parameter thermal networks (LPTNs) are widely used to predict temperature behavior in electronic systems. While these models often reproduce measured temperatures with high accuracy, a good fit alone does not guarantee that the underlying thermal parameters are uniquely determined. This case study examines the practical identifiability of a six-node LPTN for a sealed electronics enclosure under multiple operating configurations using profile likelihood analysis and Markov chain Monte Carlo (MCMC) sampling.

The estimated thermal resistances exhibited markedly different identifiability characteristics. Conductive resistances remained consistently identifiable across the configurations examined; by contrast, convective resistances were practically non-identifiable under every natural-convection condition tested. The identifiability of the plate-to-wall thermal resistance, however, varied with the operating configuration.

The results demonstrate that predictive accuracy by itself is not sufficient to justify physical interpretation of estimated thermal parameters. In this case study, covariance analysis and profile likelihood produced conflicting assessments of parameter dependence. Therefore, this case study demonstrates the need for practical identifiability methods that capture nonlinear parameter behavior.

1 Introduction

Lumped parameter thermal networks (LPTNs) approximate distributed heat transfer by utilizing networks of resistances and capacitances. They are more computationally efficient as opposed to more detailed simulations, and as such, they have become widely used for thermal management [1]. In many engineering applications, LPTNs are used by engineers to reason where heat travels and extrapolate said information for hardware diagnosis. However, often, the estimated parameters are expected to correspond to uniquely constrained physical processes as opposed to simply producing a good numerical fit.

The trouble, though, is that a model can accurately agree between predicted and measured temperatures without uniquely determining parameters. Different combinations of parameters can create nearly identical thermal responses, allowing a model to match experimental data even with poorly constrained parameters. For example, two thermal resistances in series may trade off against each other so that only their sum is constrained. This would

yield a perfect fit and extremely low root mean square error (RMSE) values, but the individual values would still be arbitrary. This class of problem also exists in systems biology and pharmacokinetics [2, 3], where the profile likelihood method was developed to address similar kinds of nonlinear compensation. Thermal modeling has been slower to adopt these tools, in part because a good RMSE is often treated as sufficient evidence of a good model.

Prior work on identifiability in thermal systems has primarily focused on how to fix the problem by modifying experimental design [4]. This case study is focused primarily on the nature of identifiability: whether changing excitation changes the kind of non-identifiability a model has, or only the degree of such non-identifiability?

This distinction has significant pragmatic consequences. If non-identifiability is purely a matter of insufficient excitation, then a better experimental design fixes it. If it is, however, inherent to the structure of the model, then no amount of additional excitation or other experimental design modifications will separate the parameters.

To investigate this, a six-node LPTN with five dynamic states was calibrated using 21 experimental runs spanning multiple heater configurations and power levels. Practical identifiability was evaluated using profile likelihood analysis, with Markov chain Monte Carlo (MCMC) posterior sampling providing complementary evidence. These two methods do not always agree.

Three distinct patterns emerge regarding identifiability. Some parameters are identifiable everywhere. One pair is non-identifiable, regardless of how strong the excitation. And one parameter is identifiable under certain heater configurations and not others, an excitation-dependent case rather than a fixed property of the model. This case study also found a case where the covariance matrix and the profile likelihood disagree over the same parameter pair, notable since covariance-based diagnostics are still the default in much of thermal modeling work.

2 Experimental Platform

2.1 Enclosure and Internal Structure

The experimental platform consists of a sealed rectangular polymer enclosure with internal dimensions of $131 \times 80.5 \times 47.7$ mm, fabricated from ABS plastic with a nominal wall thickness of approximately 4 mm. A removable, gasket-sealed lid minimizes air exchange with the ambient environment. Electrical access is provided through small wall penetrations for heater and sensor wiring.

Inside the enclosure, a 6061 aluminum plate measuring $128 \times 78 \times 3.2$ mm serves as the primary thermal distribution element. The plate is mounted with an approximately 0.5 mm clearance from the enclosure walls on all sides, which allows air to circulate within the enclosure. This clearance is particularly useful because it establishes the conductive and convective pathways represented in the lumped model. Without it, the plate-to-wall and plate-to-air resistances would not be physically distinguishable.

2.2 Heating Configuration and Excitation Conditions

Heat is supplied by resistive cement wirewound heater elements operating at discrete power levels of 5 W, 10 W, and 15 W using a 12 V DC supply. Three heater modules (H1, H2, H3) are mounted directly to the aluminum plate. Thermal excitation is varied by changing the combination of active heaters, allowing the study to evaluate how excitation influences parameter identifiability.

2.3 Temperature Instrumentation

Six 10 k Ω Negative Temperature Coefficient (NTC) thermistors ($\beta \approx 3950$) monitor temperatures at key thermal locations throughout the system:

- T_1 : On the heater surface, mounted on the central heater H2
- T_2 : On the aluminum plate, centered vertically and horizontally between H1 and H2
- T_3 : In internal air, located 2 cm above the center of the plate
- T_4 : On the interior wall surface, centered halfway between H1 and H2
- T_5 : On the exterior wall surface, directly opposite T4
- T_6 : Ambient environment

The sensor locations were selected to correspond as closely as possible to the lumped thermal nodes defined in the model in order to provide accurate spatial measurements of the dominant heat-transfer pathways. As discussed in the Results section, however, the physical T_1 measurement records the temperature of the H2 heater surface, whereas the modeled heater state represents the combined heater excitation node. This mismatch introduces a configuration-dependent limitation in the correspondence between the sensor and the modeled state.

3 Thermal Model Formulation

3.1 Assumptions

The enclosure is modeled with the following assumptions:

1. Each physical region (heater, plate, air, wall surfaces) is represented as a spatially uniform temperature node.
2. Material properties are assumed constant over the investigated temperature range.
3. Internal radiation effects are incorporated into equivalent thermal resistances.
4. Air within the enclosure is represented as a single lumped thermal node.

5. Heat transfer mechanisms are represented through equivalent thermal resistances between nodes.
6. The plate-to-wall air gap is sufficient to establish physically distinct conductive and convective pathways.

This reduces the problem from a partial differential equation (PDE) system to a coupled ordinary differential equation (ODE) system, which makes parameter estimation and identifiability analysis more manageable.

3.2 Governing Energy Balance Equations

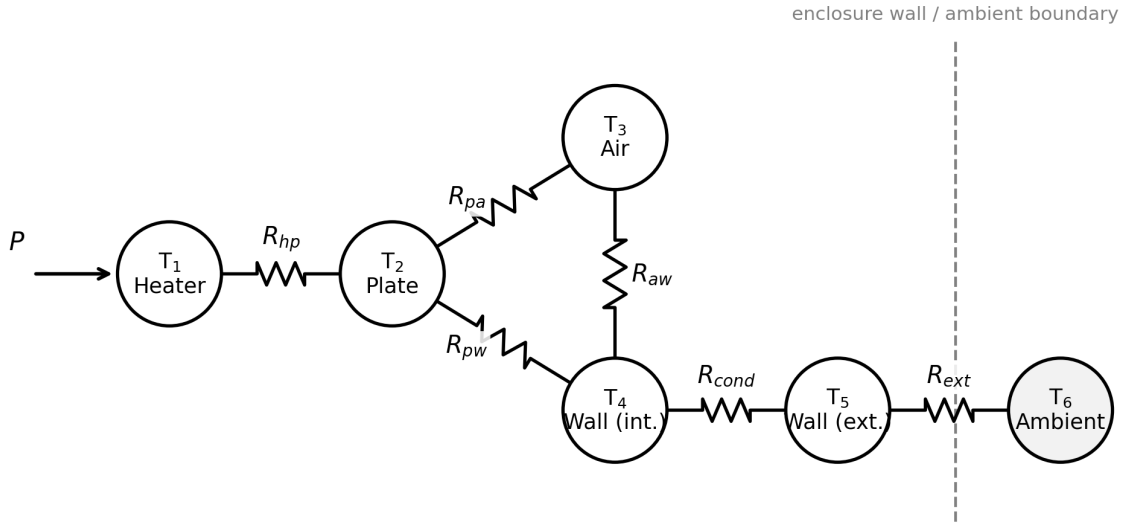


Figure 1: Representation of the six-node LPTN. Circles denote modeled temperature nodes ($T_1 - T_6$), and zigzag symbols indicate thermal resistances. Heater power input P enters at T_1 . The dashed line delineates the enclosure wall boundary to the ambient node T_6 .

The governing principle for each thermal node is the First Law of Thermodynamics. As applied to a control volume, its formula is:

$$C_i \frac{dT_i}{dt} = \dot{Q}_{in} - \dot{Q}_{out} \quad (1)$$

where $C_i = m_i c_{p,i}$ is the thermal capacitance of node i .

Heat transfer between nodes is represented using equivalent thermal resistances:

$$\dot{Q}_{ij} = \frac{T_i - T_j}{R_{ij}} \quad (2)$$

where R_{ij} represents the effective resistance associated with conduction, convection, radiation, or contact resistance between nodes.

Applying this energy balance formulation to each modeled node yields the following system:

$$C_h \frac{dT_1}{dt} = P - \frac{T_1 - T_2}{R_{hp}} \quad (3)$$

$$C_p \frac{dT_2}{dt} = \frac{T_1 - T_2}{R_{hp}} - \frac{T_2 - T_3}{R_{pa}} - \frac{T_2 - T_4}{R_{pw}} \quad (4)$$

$$C_a \frac{dT_3}{dt} = \frac{T_2 - T_3}{R_{pa}} - \frac{T_3 - T_4}{R_{aw}} \quad (5)$$

$$C_w \frac{dT_4}{dt} = \frac{T_2 - T_4}{R_{pw}} + \frac{T_3 - T_4}{R_{aw}} - \frac{T_4 - T_5}{R_{cond}} \quad (6)$$

$$C_e \frac{dT_5}{dt} = \frac{T_4 - T_5}{R_{cond}} - \frac{T_5 - T_6}{R_{ext}} \quad (7)$$

where P represents the applied heater power and the remaining parameters represent effective thermal resistances between adjacent nodes.

3.3 Thermal Resistances

The thermal network contains six effective thermal resistance parameters:

- R_{hp} : heater-to-plate contact resistance
- R_{pa} : plate-to-air resistance
- R_{aw} : air-to-wall resistance
- R_{pw} : plate-to-wall resistance
- R_{cond} : wall conduction resistance
- R_{ext} : external wall-to-ambient resistance

4 Methodology

Parameter estimation in this study assumes that jointly fitting all 21 experimental runs is valid despite their differing excitation conditions. This is reasonable because each configuration constrains a different subset of the parameter space; pooling the data allows the combined dataset to inform parameters that a single run might leave unconstrained.

The estimation procedure follows these steps:

1. All 21 experimental runs are jointly fitted using nonlinear least squares to determine $\hat{\theta}$ and establish the baseline fit quality (Section 5.1).

2. Profile likelihood scans are performed for each of the seven operating conditions (three heater configurations at 5W and 10W, plus the full H1H2H3 setup at 15W) by fixing one parameter and re-optimizing the remaining five.
3. Within each condition, profile likelihoods are evaluated independently per replicate. Parameters are classified as identifiable, partially identifiable, or non-identifiable depending on whether the profile likelihood crosses the 95% confidence limit on both sides, one side, or either. Inter-replicate discrepancies are reported explicitly to evaluate reproducibility.

4.1 Parameter Estimation

Real systems exhibit deviations from idealized analytical resistance calculations due to contact resistance variability, imperfect convection assumptions, and manufacturing tolerances. Therefore, estimation from data is necessary.

The unknown parameter vector is defined as:

$$\boldsymbol{\theta} = [R_{hp} \ R_{pa} \ R_{pw} \ R_{aw} \ R_{cond} \ R_{ext}]. \quad (8)$$

Given measured temperature data $\mathbf{T}_{exp}(t)$ and model predictions $\mathbf{T}_{model}(t; \boldsymbol{\theta})$, parameter estimation is formulated as a nonlinear least-squares optimization problem:

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \sum_{k=1}^N \|\mathbf{T}_{exp}(t_k) - \mathbf{T}_{model}(t_k; \boldsymbol{\theta})\|^2. \quad (9)$$

Parameter estimation is performed using constrained nonlinear least squares subject to the physical constraint:

$$R_i > 0. \quad (10)$$

Initial parameter values are estimated from steady-state temperature rises and equivalent thermal resistance relationships described in Section 3. These are then refined through fitting the full transient model response.

The resulting parameter estimates are evaluated using model residuals:

$$e_k = T_{exp}(t_k) - T_{model}(t_k; \hat{\boldsymbol{\theta}}). \quad (11)$$

4.2 Profile Likelihood Analysis

Unlike covariance-based uncertainty estimates, which rely on local approximations near the optimum, profile likelihood directly evaluates whether individual parameters are constrained by the available measurements.

For parameter θ_i , the profile likelihood is calculated by fixing θ_i at a series of values while optimizing all remaining parameters:

$$PL(\theta_i) = \min_{\boldsymbol{\theta}_{-i}} \chi^2(\theta_i, \boldsymbol{\theta}_{-i}), \quad (12)$$

where $\boldsymbol{\theta}_{-i}$ represents all parameters except θ_i .

A parameter is considered practically identifiable when the profile likelihood exhibits a well-defined minimum and increases beyond the confidence threshold as the parameter is varied. By contrast, a flat or asymptotic profile indicates that multiple parameter values provide statistically equivalent model predictions.

To distinguish reproducible parameter behavior from optimization-specific outcomes, identifiability is classified independently for replicate experiments.

Each parameter is classified using its profile likelihood under a given operating condition. Classifications are performed independently for each replicate. When discrepancies arise between replicates, they are explicitly reported rather than omitted. This approach prevents pooled datasets from masking run-specific or configuration-dependent effects, which allows for parameter identifiability to be evaluated at the level of individual excitation conditions and single runs.

Profile likelihood thresholds, rather than covariance approximations, are used to obtain confidence intervals in this study. The 95% confidence interval is defined by the parameter values that satisfy:

$$\chi^2(\theta_i) - \chi_{\min}^2 \leq \chi_{0.95}^2 \quad (13)$$

This chi squared threshold ($\chi_{0.95}^2 = 3.84$) follows from a one-sided chi-squared distribution with one degree of freedom, corresponding to profiling a single parameter. The value defines the boundary at which a candidate parameter value can be rejected at 95% confidence. The value is standard practice in profile-likelihood analysis [2].

4.3 Markov Chain Monte Carlo Analysis

Posterior distributions are sampled using the affine-invariant ensemble MCMC method implemented in emcee [5], using 32 walkers initialized in a small Gaussian ball around the least-squares optimum, with flat priors over physically admissible resistance values (10^{-4} to 10^3 K/W). A short 500-step diagnostic chain was run first to verify walker mixing and estimate the autocorrelation time; the production chain was then run for 8000 steps per walker. Burn-in and thinning were selected from the estimated autocorrelation time τ , discarding the first $3 \max(\tau)$ steps and thinning by $0.5 \min(\tau)$, following standard emcee convergence practice.

Marginal and joint posterior distributions were then examined to compare inferred parameter dependencies with those obtained from profile likelihood analysis, allowing covariance-based interpretations to be evaluated against nonlinear parameter compensation.

5 Results

Table 1 summarizes the identifiability outcome for each of the six thermal resistances. The remainder of this section presents the evidence supporting each classification.

5.1 Model Fit

The model reproduced the four spatially distributed states (T2–T5) with good agreement across all 21 experimental runs. Figure 2 shows a representative fit for Run 19 (15 W,

Table 1: Summary of practical identifiability for the six thermal resistances.

Resistance	Identifiability	Simple meaning
R_{hp}	Identifiable	Its value could be estimated
R_{cond}	Identifiable	Its value could be estimated
R_{ext}	Identifiable	Its value could be estimated
R_{pa}	Not identifiable	It could trade off with R_{aw}
R_{aw}	Not identifiable	It could trade off with R_{pa}
R_{pw}	Depends on heater setup	Some experiments could estimate it; others could not

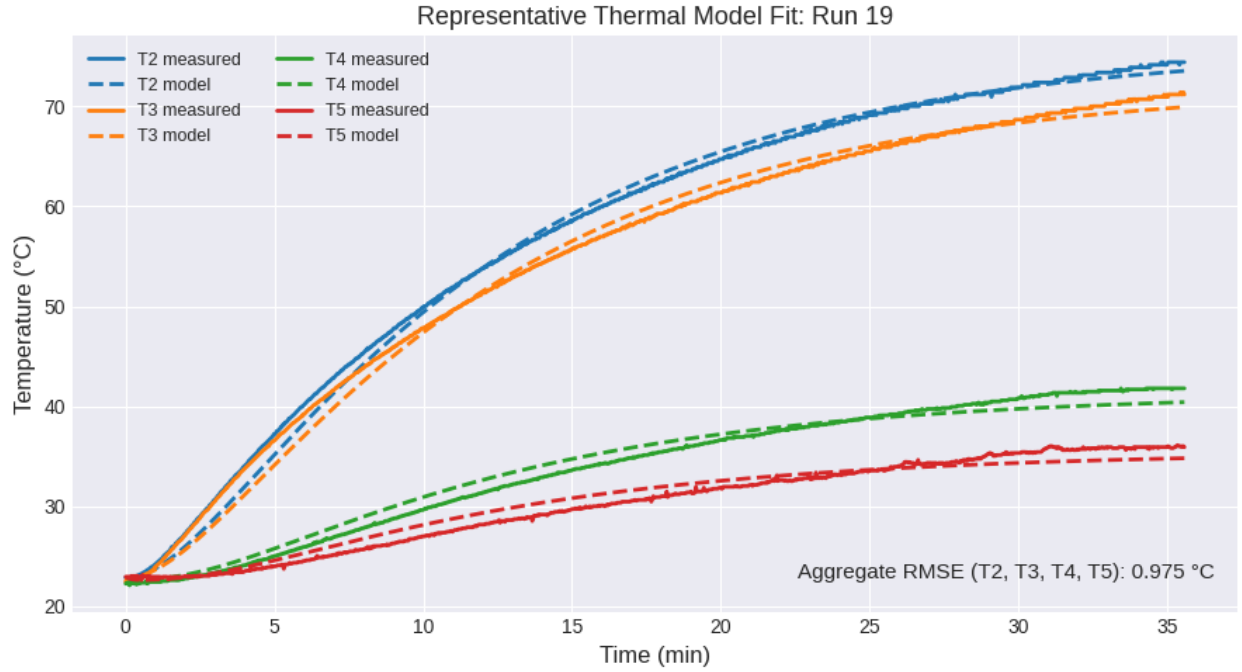


Figure 2: Measured and simulated temperatures for Run 19 (15 W, H1H2H3 configuration). The model reproduces the transient responses of nodes T2–T5 with an aggregate RMSE of 0.975 °C. Solid curves denote measured temperatures and dashed curves denote model predictions.

H1H2H3), in which the model tracks the measured transient and steady-state response of each node closely. The largest error observed across all nodes and operating conditions was 1.54°C at T_3 under the H2H3 configuration at 10 W, placing the aggregate prediction error for the spatially distributed states below 2°C .

The largest residual occurred at T_3 , the low-capacitance free-air node, whose temperature is governed primarily by convective heat transfer. This makes it more sensitive both to unmodeled air motion and to uncertainty in the convective resistance parameters. The solid nodes, by contrast, are dominated by conduction and buffered by the thermal mass of the enclosure, making their temperatures substantially less sensitive to these effects.

Node T_1 was excluded from the aggregate fit metric because the modeled state represents the combined heater excitation, whereas the physical T_1 measurement is obtained from a thermistor attached to the H2 heater body. These quantities coincide only when H2 is the sole active heater. Under all other heater configurations, the residual reflects both model error and a configuration-dependent mismatch between the modeled state and the physical measurement. For this reason, T_1 residuals are reported separately (Appendix B, Table 2), while aggregate fit statistics are based on T2–T5.

5.2 Identifiable Parameters

Three thermal resistance parameters— R_{hp} , R_{cond} , and R_{ext} —were practically identifiable over the operating conditions investigated in this study.

The heater-to-plate resistance, R_{hp} , is primarily constrained by the plate temperature measurement, T_2 . Heat transfer across this interface governs both the transient and steady-state behavior of the aluminum plate, producing a response that cannot be replicated by downstream conduction or external heat loss.

The conduction resistance, R_{cond} , is determined by the temperature difference between the inner and outer wall measurements (T_4 and T_5), while the external resistance, R_{ext} , is constrained by the gradient between the exterior wall (T_5) and the ambient environment (T_6).

This separation is reflected in the well-defined profile likelihoods across all three parameters (Figure 3). The covariance matrix reveals only weak correlation between R_{hp} and the remaining resistances ($r = 0.10$ with R_{cond} , and $r = 0.17$ with R_{ext}). Conversely, R_{cond} and R_{ext} exhibit a stronger negative correlation ($r = -0.72$), which aligns with their series combination in the steady-state model reduction (Appendix A).

Each of the three identifiable parameters exhibits a profile likelihood with a clear minimum and finite confidence bounds on both sides of the optimum. The resulting 95% confidence intervals are $R_{hp} = [6.08, 6.45] \text{ K/W}$, $R_{cond} = [0.380, 0.406] \text{ K/W}$, and $R_{ext} = [0.833, 0.885] \text{ K/W}$, obtained after confirming convergence of the profile likelihood calculations.

The replicate consistency criterion described in Section 4 further showed that these parameters remained identifiable across repeated experiments under the same operating conditions. Their identifiability therefore reflects a reproducible property of the experimental system rather than an artifact of a particular optimization run or pooled dataset.

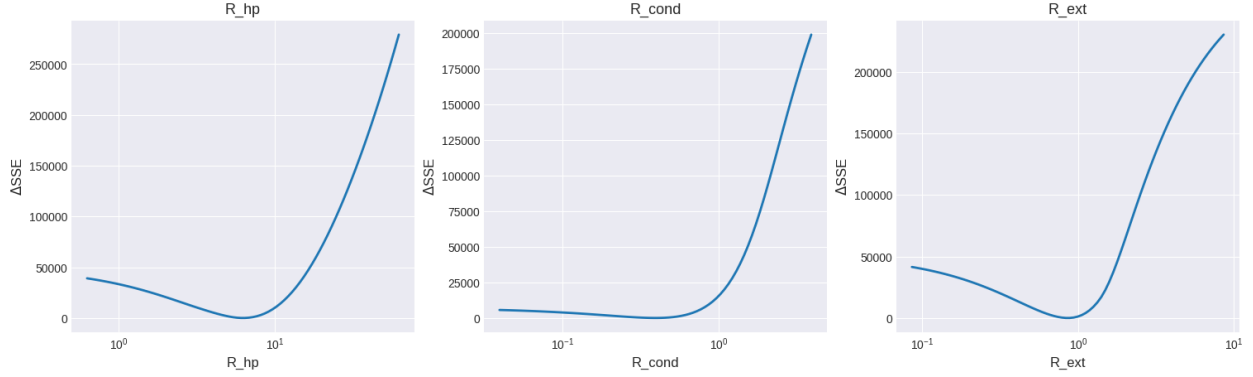


Figure 3: Profile likelihoods, expressed here as increases in the objective function (Δ Sum of Squared Errors, SSE), for the conductive resistances R_{hp} , R_{cond} , and R_{ext} from the representative Run 19 fit. At each fixed parameter value, the remaining parameters were re-optimized to their conditional optimum. Each profile exhibits a well-defined minimum, with the objective increasing as the parameter is displaced from its optimum, consistent with practical identifiability of these parameters under the tested experimental conditions. Contrasting profile behavior associated with persistent practical non-identifiability of the convective resistances and the configuration-dependent identifiability of R_{pw} is shown in Figures 3 and 4.

5.3 Convective Collinearity

In contrast to the conductive parameters discussed in Section 5.2, the convective resistances were practically non-identifiable under natural convection. The dominant feature of this behavior is the near-perfect collinearity between the plate-to-air resistance, R_{pa} , and the air-to-wall resistance, R_{aw} . The measurements therefore constrain only their combined thermal effect rather than the individual resistance values.

This degeneracy is evident in the profile likelihoods. Whereas the conductive parameters exhibit well-defined minima with finite confidence intervals (Section 5.2), the profiles for R_{pa} and R_{aw} remain flat or asymptotic over a broad range of parameter values (Figure 4). Large changes in one resistance are offset by compensating changes in the other, producing little increase in the objective function despite the strong predictive agreement reported in Section 5.1.

A notable finding is that the covariance approximation does not fully capture the local geometry of the parameter coupling. In one representative case, Run 16 (H25W)’s marginal correlation matrix indicates a moderate negative correlation between R_{pa} and R_{aw} ($r = -0.52$), whereas the corresponding profile likelihood is distinctly nonlinear and asymmetric. Near the optimum, the profile exhibits a weakly negative slope on one branch (approximately -0.55) before turning sharply positive on the other (approximately $+8.88$). Previous studies have shown that Fisher information and covariance-based methods can misrepresent parameter dependencies when practical non-identifiability is strongly nonlinear [2, 3]. The present results provide an example within a lumped thermal network in which the covariance structure obscures important nonlinear features of the profile-likelihood geometry.

This degeneracy persists across all natural-convection experiments, including the highest power level of 15 W, which is consistent with a limitation imposed by the thermal network

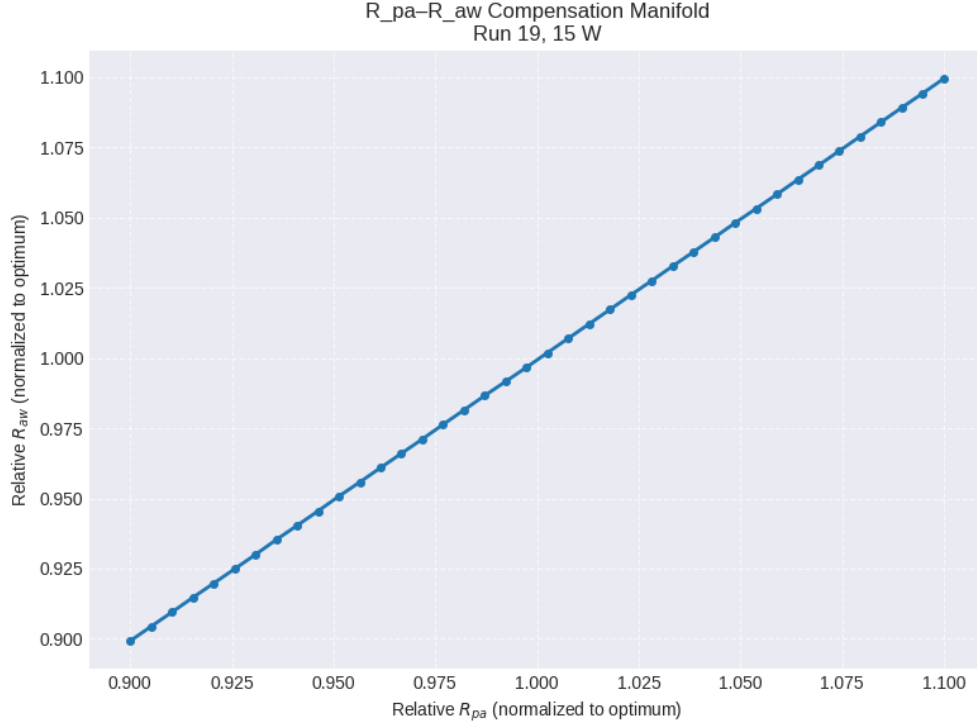


Figure 4: Compensation manifold for the coupled convective resistances R_{pa} and R_{aw} in Run 19 (15 W, H1H2H3). The conditional profile-likelihood trajectory is shown in coordinates normalized to the unconstrained fitted values of each parameter. The approximately straight, positively sloped trajectory over the range 0.90–1.10 times the respective fitted values demonstrates an approximately 1:1 degenerate parameter manifold between R_{pa} and R_{aw} .

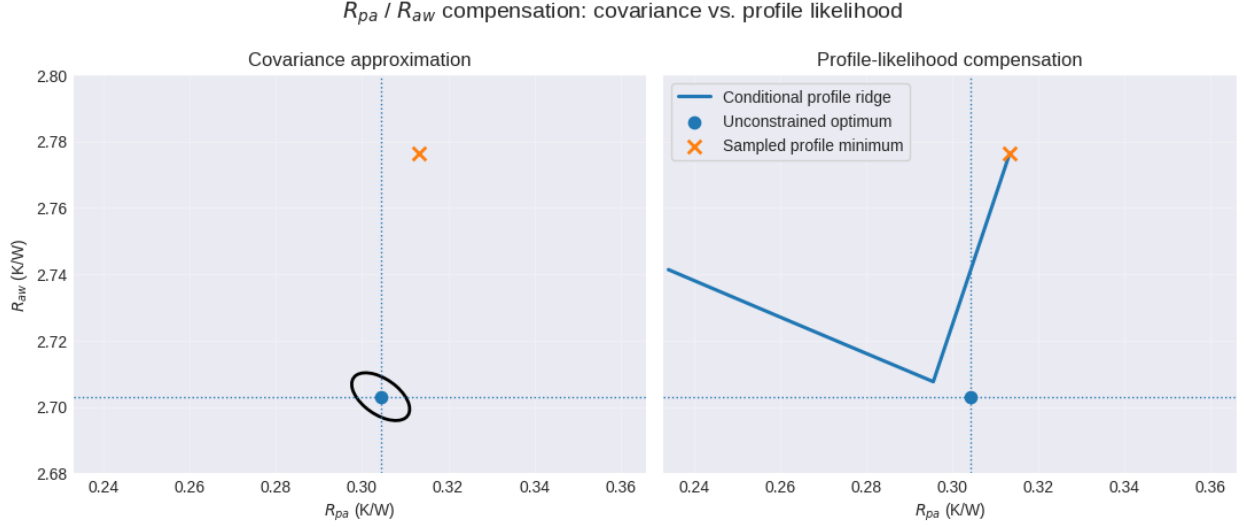


Figure 5: Covariance-ellipse approximation versus conditional profile-likelihood compensation for R_{pa} and R_{aw} in Run 16 (H2, 5W). Unlike the approximately linear manifold in Run 19 (Figure 4), the profile ridge shown here comes from Run 16 and captures curvature near the optimum that is not visible at Run 19’s coarser sampling. The left panel shows the 1- σ covariance ellipse derived from the Jacobian-based covariance matrix, centered at the unconstrained fit optimum $(R_{pa}, R_{aw}) = (0.3043, 2.7031) K/W$. The covariance structure indicates a negative marginal association between the two parameters ($r = -0.523$). The right panel shows the corresponding conditional profile-likelihood ridge obtained by re-optimizing the remaining parameters at each sampled R_{aw} value. The profile ridge is asymmetric about the optimum, with a negatively sloped left branch and a positively sloped right branch, producing a kink near the optimum that is not captured by the covariance approximation. The circle marks the unconstrained fit optimum, whereas the X marks the sampled profile minimum at $R_{aw}=0.3133 K/W$; the difference reflects the finite sampling of the profile grid rather than a second fitted optimum.

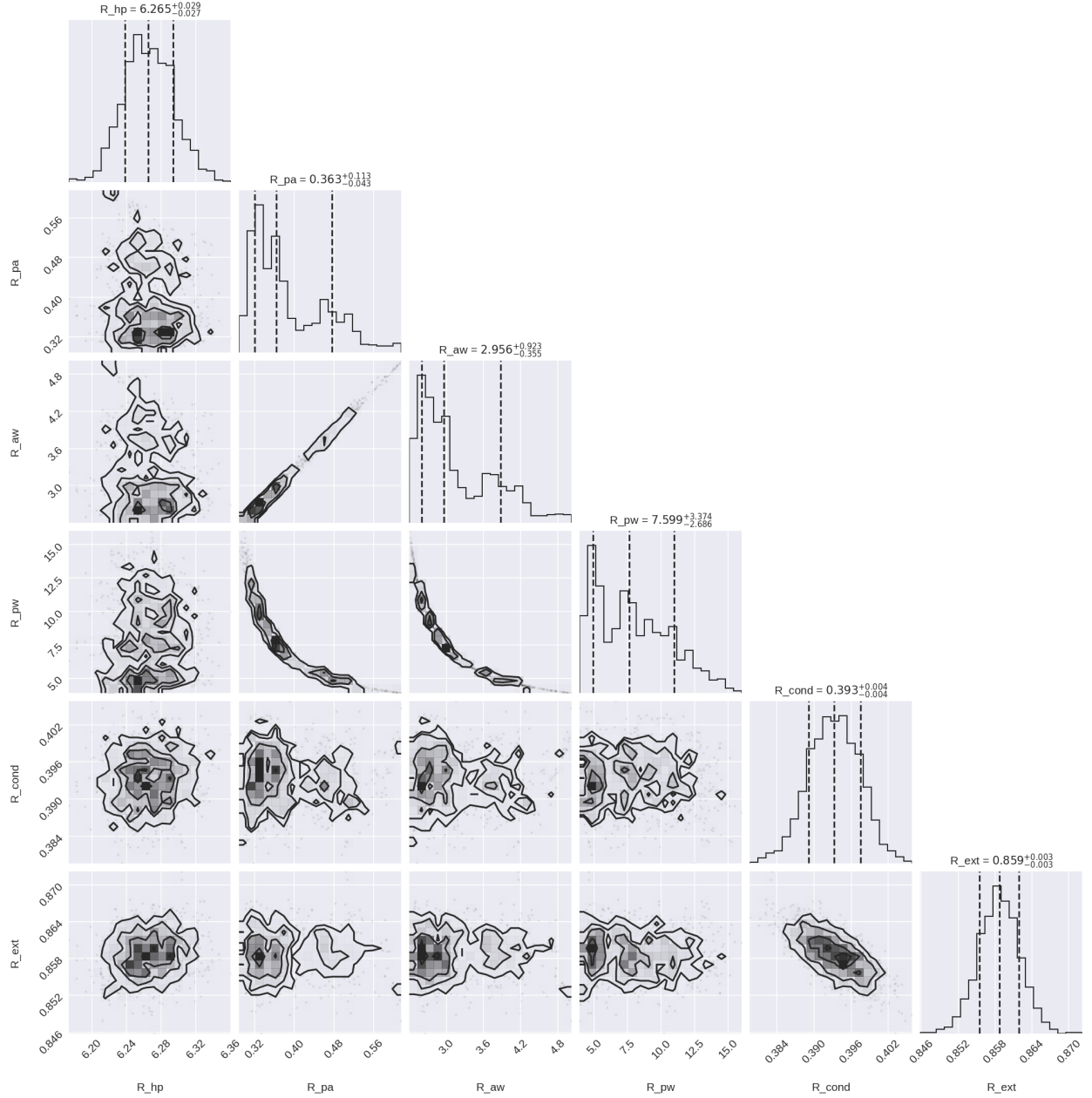


Figure 6: The joint posterior for R_{pa} and R_{aw} forms a narrow diagonal ridge consistent with the compensation manifold observed in the profile likelihood analysis (Figure 4), providing complementary evidence from an independent estimation method.

architecture rather than weak excitation alone. The resistances R_{pa} and R_{aw} govern consecutive stages of the same heat-transfer pathway through a lumped air node, leaving the available measurements sensitive only to their combined effect. Under this model structure, additional excitation, reduced measurement noise, or higher-quality sensors cannot independently constrain the two parameters because the ambiguity is structural to the model rather than informational in the data. Resolving it would require a revised thermal network architecture that separates the air pathway into independently observable thermal regions. Independent support for this interpretation is provided by the MCMC corner plot (Figure 6), where the joint posterior for R_{pa} and R_{aw} forms the same narrow diagonal ridge predicted by the profile likelihood.

The convective resistance estimates should therefore be interpreted as effective parameters of the lumped thermal model rather than direct measurements of isolated convective coefficients. Because T_3 samples a localized region within a non-uniform air volume, the estimated values of R_{pa} and R_{aw} may also absorb spatial temperature variations that are not explicitly represented in the model. This limitation does not affect either the fit-quality or identifiability conclusions presented here, which concern the ability of the model structure to reproduce the measured thermal response and constrain its parameters. Instead, it reinforces an important distinction: practical identifiability describes whether a parameter can be estimated within a model, not whether that parameter necessarily represents a uniquely separable physical mechanism.

Taken together with Sections 5.1 and 5.2, these findings show that excellent predictive performance does not imply parameter identifiability. The same model that achieves sub-2°C prediction error and uniquely identifies the conductive resistances also contains a persistently practically non-identifiable convective subnetwork in natural convective conditions.

5.4 Plate–Wall Contingency

The plate-to-wall thermal resistance, R_{pw} , exhibited a different pattern. Unlike the consistently identifiable conductive parameters (Section 5.2) or the persistently non-identifiable convective resistances (Section 5.3), the practical identifiability of R_{pw} depended on the experimental configuration. Across the operating conditions examined in this study, R_{pw} was practically non-identifiable under natural convection in every H2-on experiment, encompassing all four heater configurations at each of the three power levels investigated (12 runs total).

The converse conclusion cannot be drawn. All verified H2-off experiments produced profile likelihoods consistent with practical identifiability, but H2-off cannot be claimed to guarantee identifiability. One H2-off experiment (Run 18) contained an unverified anomaly in the T_2 measurement that prevents an unambiguous classification. The evidence therefore supports the asymmetric conclusion that H2-off is necessary but not sufficient for practical identifiability of R_{pw} within the tested operating regime. No attempt is made to explain the anomalous observation beyond what is supported by the available data.

The profile likelihoods clearly distinguish these two cases. In every H2-on experiment, the profile for R_{pw} remained flat or asymptotic over the explored parameter range, indicating that substantial changes in the parameter produced little increase in the objective function. The verified H2-off experiments, by contrast, exhibited well-defined likelihood minima with finite

confidence bounds, demonstrating that R_{pw} was constrained by the available measurements. The difference therefore arises from the information content of the experimental configuration rather than from optimization performance.

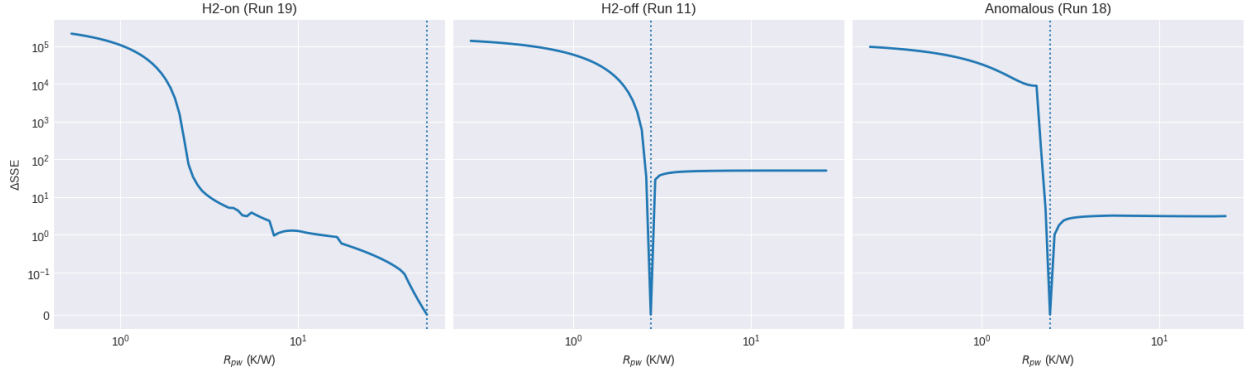


Figure 7: Profile objective for R_{pw} across representative natural-convection runs, showing the dependence of practical parameter constraint on excitation configuration. Across the full run set, all 12 H2-on runs exhibited effectively flat high- R_{pw} tails, while two of three H2-off runs retained bounded increases in the objective and Run 18 represented an anomalous case. For the representative runs shown, the corresponding high- R_{pw} tails represent RMS-equivalent temperature penalties of approximately 0.004 °C (Run 19), 0.024 °C (Run 18), and 0.098 °C (Run 11), spanning approximately a 20–25-fold range. These values express the profile-objective differences directly in temperature-error units.

To verify that this pattern was reproducible, identifiability was classified independently for each replicate using the criterion established in Section 4. Rather than pooling datasets, each of the three replicates within an operating condition was evaluated separately. Practical non-identifiability was observed consistently across all H2-on replicates, whereas pooling the data would have obscured the configuration-dependent behavior.

No further mechanism is inferred beyond what is supported by the profile likelihood analysis.

Within the experimental domain considered here, the practical identifiability of R_{pw} is governed not only by parameter sensitivity but also by whether the excitation condition provides sufficient information to constrain the parameter.

6 Discussion

The results show that changing the excitation regime reshaped the parameter identifiability landscape rather than simply improving parameter estimation. Predictions stayed accurate across all 21 runs. Which parameters were actually constrained by those predictions, however, changed with excitation. This distinction is significant because predictive accuracy alone is often taken as evidence that fitted parameters are trustworthy.

Three distinct patterns of practical identifiability emerged. The conductive resistances (R_{hp} , R_{cond} , and R_{ext}) remained identifiable throughout the experimental regime. The convective resistances (R_{pa} and R_{aw}), by contrast, remained practically non-identifiable under

natural convection despite broader excitation. Additionally, R_{pw} occupied an intermediate case in which identifiability depended on the heater configuration.

The large residual at the modeled heater node initially suggested that R_{hp} might be poorly identifiable. The governing equations and profile likelihoods showed otherwise: R_{hp} is constrained primarily through the plate temperature measurement, not the heater-state measurement. Residual magnitude alone was an unreliable indicator of identifiability in this study.

This is consistent with prior work showing that accurate predictions do not guarantee identifiable parameters [2, 3]. The coupled convective resistances are a clear example. Covariance analysis suggested a moderate negative correlation between R_{pa} and R_{aw} ; the profile likelihood instead revealed a positively sloped, asymmetric compensation manifold. Local covariance measures can gloss over such kinds of dependency when non-identifiability is strongly nonlinear [3].

These conclusions are limited to the operating conditions examined here and do not imply a universal relationship between excitation design and identifiability in lumped thermal networks. Excitation should instead be viewed as a factor that can change the structure of the estimation problem itself, not merely the precision of the resulting estimates.

7 Limitations

7.1 Lumped-Model Simplification

The present model represents the heater, aluminum plate, internal air, and enclosure walls as spatially uniform temperature nodes. This reduces the distributed heat equation to a coupled system of ordinary differential equations, which is what makes parameter estimation tractable. It also neglects temperature gradients within individual components; the aluminum plate may exhibit localized heating near the heater footprints, and the air within the enclosure is not likely to remain perfectly isothermal under natural convection.

7.2 Contact Resistance Uncertainty

The heater-to-plate contact resistance, R_{hp} , cannot be determined reliably from material properties and geometry alone. Unlike bulk conduction resistances, its value depends on interface conditions such as surface roughness, contact pressure, and the presence and distribution of thermal interface material, so it must be estimated from experimental data rather than calculated previously. Non-uniform mounting pressure or inconsistent application of thermal paste can introduce run-to-run variation in these conditions, which the deterministic model does not explicitly represent.

Although R_{hp} was practically identifiable under the conditions investigated here, the estimated value is an effective interface parameter, not an intrinsic material property. It reflects the specific assembly conditions of this apparatus, and may not transfer to systems with different mounting procedures.

7.3 Measurement and Sensor Uncertainty

Temperature measurements were acquired using 10 k Ω NTC thermistors in a voltage-divider configuration, and were sampled with a 10-bit analog-to-digital converter. Thermistor accuracy is limited by manufacturing tolerances in the nominal resistance and the β parameter, and ADC voltage quantization adds discrete measurement error that grows more pronounced where thermistor sensitivity is lower.

Each thermistor also measures a single physical location, while the lumped model represents each component with a spatially averaged temperature. This mismatch is most significant for the enclosure air, where one sensor stands in for the internal air volume, which is unlikely to be isothermal under natural convection. These measurement uncertainties propagate into the estimated resistances and their confidence bounds.

8 Future Work

Future work should investigate excitation strategies and thermal network architectures that improve separation of coupled heat transfer pathways. The convective collinearity identified in Section 5.3 arises because R_{pa} and R_{aw} act on a single lumped air node; separating them would require either spatially resolved air modeling or an additional internal temperature measurement placed between the plate and the wall. A similar approach could test whether the R_{pw} contingency observed in Section 5.4 depends on heater placement specifically, or on some other feature of the H2-on configurations.

9 Conclusion

This study demonstrated that accurate thermal prediction can coexist with substantially different levels of parameter identifiability within the same lumped thermal model. Three parameters were identifiable everywhere, two were practically non-identifiable under tested natural-convection conditions, and one was identifiable only when a specific heater was inactive. When LPTNs are used to infer physical quantities rather than only predict temperatures, identifiability cannot be assumed from fit quality alone. It has to be verified under the configurations the model will actually be used in.

A Closed Form Steady-State Reduction

At steady state:

$$\frac{dT_i}{dt} = 0 \tag{14}$$

and the governing equations reduce to algebraic heat-flow balances.

For the heater node:

$$0 = P - \frac{T_1 - T_2}{R_{hp}} \tag{15}$$

which gives:

$$T_1 = T_2 + PR_{hp} \quad (16)$$

The internal air pathway contains two consecutive thermal resistances, R_{pa} and R_{aw} . Because the same steady-state heat flow passes through both resistances:

$$\frac{T_2 - T_3}{R_{pa}} = \frac{T_3 - T_4}{R_{aw}} \quad (17)$$

the air-mediated pathway is governed by their combined resistance:

$$R_{air} = R_{pa} + R_{aw} \quad (18)$$

Similarly, the wall conduction and external heat-loss pathways combine in series:

$$R_{wall} = R_{cond} + R_{ext} \quad (19)$$

The steady-state response therefore depends on the equivalent plate-to-wall resistance:

$$R_{2 \rightarrow 4} = \left(\frac{1}{R_{pw}} + \frac{1}{R_{pa} + R_{aw}} \right)^{-1} \quad (20)$$

and the total plate-to-ambient resistance:

$$R_{2 \rightarrow 6} = \left(\frac{1}{R_{pw}} + \frac{1}{R_{pa} + R_{aw}} \right)^{-1} + R_{cond} + R_{ext} \quad (21)$$

giving:

$$T_2 - T_6 = PR_{2 \rightarrow 6} \quad (22)$$

Equation 18 shows that R_{pa} and R_{aw} enter the steady-state response only through their sum. This is the algebraic origin of the collinearity between them identified in Section 5.3. No steady-state measurement can constrain the two resistances separately, only their combined value.

B T1 Residuals by Heater Configuration

Because T_1 does not consistently correspond to the same physical node represented by the modeled T_1 state across heater configurations, it was excluded from the aggregate fit statistics reported in Section 5.1. The residuals are nevertheless reported here to make the effect of this sensor-to-node mismatch transparent.

Table 2: T1 root-mean-square error (RMSE) by heater configuration. The three H2-only runs are reported individually because the physical T_1 measurement coincides with the modeled heater-excitation node in this configuration. For all other configurations, T1 RMSE is reported as the mean across three replicate runs.

Run	Configuration	Power (W)	T1 RMSE ($^{\circ}\text{C}$)
2	H2	5	15.42
9	H2	5	2.71
16	H2	5	8.81
–	H1 (mean, $n = 3$)	5	30.59
–	H3 (mean, $n = 3$)	5	36.54
–	H1H2 (mean, $n = 3$)	10	18.96
–	H2H3 (mean, $n = 3$)	10	18.84
–	H1H3 (mean, $n = 3$)	10	70.38
–	H1H2H3 (mean, $n = 3$)	15	59.46

T1 RMSE is reported individually for the three H2-only replicates, in which the physical T1 measurement coincides with the modeled heater-excitation node, and as the mean across three replicates for all other configurations, in which T1 reflects both model error and a configuration-dependent mismatch between the physical sensor and the modeled state. T1 is excluded from the aggregate fit statistics reported in Section 5.1 for this reason.

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